

Chang Lv

Associate Professor

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Education

- 2018–Now **Associate Professor**, *Institute of Information Engineering, CAS, Beijing, China.*
- 2015–2018 **Assistant Professor**, *Institute of Information Engineering, CAS, Beijing, China.*
- 2010–2015 **Ph.D.**, *Academy of Mathematics and Systems Science, CAS, Beijing, China,*
- Degree major: Applied Mathematics, Adviser: Zhexian Wan and Yingpu Deng.
 - Thesis: Primality tests, ring class fields and diophantine equations, 2015
- 2006–2010 **B.S.**, *Nankai University, Tianjin, China,*
- Degree major: Mathematics and Applied Mathematics.

Area of research

I am interested in **number theory**, **arithmetic geometry** and **cryptography**.

Fundings

- 01/2018-12/2020 Existence of integral points on quadratic norm form equations over global fields and applications, NNSF of China (Grants No. 11701552)

Publications

- [DL15] Yingpu Deng and Chang Lv, *Primality test for numbers of the form $Ap^n + w_n$* , *Journal of Discrete Algorithms* **33** (2015), 81–92.
- [LD15] Chang Lv and Yingpu Deng, *On orders in number fields: Picard groups, ring class fields and applications*, *Science China Mathematics* **58** (2015), no. 8, 1627–1638.
- [LL17] Chang Lv and Jianing Li, *On the non-existence of certain classes of generalized bent functions*, *IEEE Transactions on Information Theory* **63** (2017), no. 1, 738–746.
- [LLP24] Xiao Li, Chang Lv, and Zhizhong Pan, *A generalized MSST algorithm for counting points of elliptic curves over $\mathbb{F}_{p^n}$* , *J. Syst. Sci. Complex.* **37** (2024), no. 4, 1738–1754. MR 4757054
- [LSD18] Chang Lv, Junchao Shentu, and Yingpu Deng, *On the integral representation of binary quadratic forms and the Artin condition*, *Tokyo J. Math.* **41** (2018), no. 2, 371–384. MR 3908800
- [Lv17] Chang Lv, *On the non-existence of certain classes of perfect p -ary sequences and perfect almost p -ary sequences*, *IEEE Transactions on Information Theory* **63** (2017), no. 8, 5350–5359.

- [Lv20a] ———, *Diophantine equations defined by binary quadratic forms over rational function fields*, *Acta Arith.* **196** (2020), no. 1, 35–51. MR 4129970
- [Lv20b] ———, *A note on the Brauer group and the Brauer-Manin set of a product*, *Bull. Lond. Math. Soc.* **52** (2020), no. 5, 932–941. MR 4171413
- [Lv21] ———, *The second descent obstruction and gerbes*, arXiv preprint arXiv:2009.06915v2 (2021), 1–31.
- [Lv22] ———, *On commutative diagrams consisting of low term exact sequences*, arXiv preprint arXiv:2112.14386v2 (2022), 1–9.
- [Lv26] ———, *Cohomological descent for obstructions to local-global principle*, arXiv preprint arXiv:2603.23288 (2026), 1–28.
- [LW23] Chang Lv and Han Wu, *The Brauer-Manin obstruction on algebraic stacks*, arXiv preprint arXiv:2306.14426v2 (2023), 1–16.
- [LW26] ———, *Comparing local-global obstructions on algebraic stacks*, in preparation (2026), 1–10.
- [LZ25] Chang Lv and Yuqing Zhu, *Nonexistence of generalized bent functions and the quadratic norm form equations*, *Des. Codes Cryptogr.* **93** (2025), no. 7, 2543–2559. MR 4940228
- [WL24a] Han Wu and Chang Lv, *Genus one half stacky curves violating the local-global principle*, *Journal de Théorie des Nombres de Bordeaux* **36** (2024), no. 3, 835–841.
- [WL24b] ———, *Iterated descent obstructions for algebraic stacks*, arXiv preprint arXiv:2409.20068 (2024), 1–9.
- [YL18] Hairong Yi and Chang Lv, *On ring class fields of number rings*, arXiv preprint arXiv:1810.04810 (2018), 1–17.

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